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Problem Set: Inference

Question 1:

$\bar{X} = 52 \quad S = 10 \quad n = 100$

$SE = 10/\sqrt{100} = 1$

95% confidence interval = $52 \pm 1.96(1) = (50.04, 53.96)$

$Z = \frac{52-50}{1} = 2$

90%: $2 > 1.645$, Reject H_0

95%: $2 > 1.96$, Reject H_0

99%: $2 < 2.576$, fail to reject H_0

Question 2:

$Z = \frac{3.1}{1.3} = 2.38 \quad H_0: S = 0 \quad H_1: S \neq 0$

90% CI: $3.1 \pm 1.645(1.3) = (0.96, 5.24)$, Reject H_0

95% CI: $3.1 \pm 1.96(1.3) = (0.55, 5.65)$, Reject H_0

99% CI: $3.1 \pm 2.576(1.3) = (-0.25, 6.45)$, Fail to reject H_0

The null hypothesis is rejected when zero is not in the confidence interval.

Question 3:

$SE = \frac{3}{\sqrt{36}} = 0.5 \quad Z = \frac{11.5-12}{0.5} = -1 \quad H_0: \mu = 12 \quad H_1: \mu \neq 12$

95% CI: $11.5 \pm 1.96(0.5) = (10.52, 12.48)$, fail to reject H_0

12 is plausible because it is in the confidence interval.

Question 4:

$\bar{X}_A - \bar{X}_B = 25-22 = 3 \quad SE = \sqrt{(8^2/100 + 7^2/100)} = \sqrt{1.13} = 1.06$

$Z = 3/1.06 = 2.83$

95% CI: $3 \pm 1.96(1.06) = (0.92, 5.08)$, reject the H_0

Question 5:

$\bar{X}_A - \bar{X}_B = 14-12 = 2 \quad SE = \sqrt{(6^2/81 + 5^2/64)} = \sqrt{0.835} = 0.914 \quad Z = \frac{2}{0.914} = 2.19$

Question 6:

$X_A - X_B = 3.2 - 2.8 = 0.4$

$SE = \sqrt{(2.25/49 + 1.96/64)} = 0.277$

$Z = 4/0.277 = 1.45$

95% CI: $0.4 \pm 1.96(0.277) = (-0.14, 0.94)$

fail to reject the null hypothesis because zero is in the CI

and $1.45 < 1.96$